GREEN TECHNOLOGY

Green Technology is the development and application of products, equipment, and systems used to conserve the natural environment and resources, which minimize and reduces the negative impact of human activities.

Why green technology

Green technology is not a new technology; instead, it has picked momentum with the introduction of sustainable development initiatives because it is an environmentally friendly technology and hence known as environmental technology or clean technology. It uses innovative methods to create environmentally friendly products. Moreover, Green technology is known as sustainable technology as it takes into account the long- and short-term impact something has on the environment.

The need for green technology arises because natural resources are declining and pollution has increased due to the abundant use of non-renewable sources. Green products and creativities are by definition, environmentally friendly, energy efficiency, recycling, health and safety concerns, renewable resources, and more all go into the making of a green product or technology.

Products, systems or equipment based on green technology satisfy the following features or characteristics.

- It should reduce or minimize the degradation of the natural environment around us. It should have zero or minimum emission of greenhouse gases.
- It should be safe to use and should promote a healthy environment for all the forms of life including humans, birds, animals, etc.
- It should help in the conservation of energy as well as natural resources such as solar, water, wind, etc.
- It should promote the use of renewable resources.

Major applications of green technology

i. **Energy sector**

South Africa is the African continent's leader when it comes to renewable energy capacity. It is home to several solar power plants, including the 96 MW Jasper plant, a photovoltaic facility with 325,360 panels that can supply power to 80,000 homes. When it comes to wind power, onshore installations in South Africa amounted to 2,085 MW in 2018, according to the Global Wind Energy Council. The South African Wind Energy Association states that wind power accounts for 52% of the country's renewable energy power supply, while the average size of a South African wind power facility is 93.5 MW.

ii. **Building sector**

The applications of green technology solutions within the construction industry are fast improving. For instance, buildings that incorporate solar energy systems use up to 70% less electricity than conventional systems, according to the U.S Department of Energy. Also, green technology helps reduce a business' carbon footprint, reduces waste, conserves water and consumes reduced energy as compared to traditional technologies.

iii. Transport sector

Here, it is commonly referred to as Green Transportation or Sustainable Transportation which comprises those modes of transportation that do not depend on diminishing natural resources like fossil fuels. These transportation modes rely on renewable energy sources. They also have a very low impact on the environment as these modes produce minimal or no greenhouse gas emission.

Strengths / advantages of green technology

The world has a fixed amount of natural resources, some of which have already been depleted or ruined. For example, household batteries and electronics often contain dangerous chemicals that pollute our environments. Green technology offers us the best hope to counteract the effects of climate change and pollution. It would be productive to list all the benefits of green technologies in a point wise manner with examples.

i. **It does not emit or emit less harmful substances into the environment**. For instance; the solar cell, which directly converts energy from natural light into electrical energy via the process of photovoltaic. Generating electricity from solar energy equates to less consumption of fossil fuels, as well as the reduction of pollution and greenhouse gas emissions.

ii. **Rejuvenating Ecosystems**. Green or Clean technology is also being used to breathe life into ecosystems that have sustained a lot of damage due to human involvement. Through the use of this technology, trees are replanted, waste is managed and recycled. This ensures that the affected ecosystem can start again, and this time remains conserved. This helps to ensure that a lot of plant and animal species don't go extinct.

iii. **It has become popular as consumers of technology** are becoming more environmentally conscious. This will give benefits to investors in the long run in certain areas. A country Like Uganda has favorable solar irradiation of 1,825 kWh/m² to 2,500 kWh/m² per year. Small solar applications are often used in rural electrification projects such as Solar Home Systems or solar water heating. Over 30,000 solar PV systems have already been installed in rural areas. This brings more applicability of renewable energy technologies as well as bringing profits to the investors. This had major impacts on rural electrification as well as investment benefits to the private sector.

iv. **It requires less cost for maintenance**. This reduces operating costs and hence overall cost in the long run. For instance, Solar panels generally require very little maintenance. They are very durable and should last around 25-30 years with no maintenance. The only maintenance one needs to perform is to wash them clean of dirt and dust two to four times a year. This basic cleaning routine ensures that the sun can shine brightly on the panel, maximizing the amount of light available to turn into electrical power. It is the contrast to oil refineries or other non-renewable energy sources that require too much maintenance and constantly, that if not done can lead to major destructive accidents.

v. As it uses renewable natural resources and hence we will never run out of vital resources such as water and electricity. With renewable energy sources, such as wind power and solar power such problems will be less likely to encounter the problem of fuel getting vanished.

Energy is being conserved through the use of such technology. Alternatives to devices that use a lot of electricity or fuel are being introduced to the public. The use of electric cars is on the rise, especially in the UK. People are using environment-friendly devices and appliances. While installation of such devices, namely solar panels, might be expensive for some people the benefits it offers with regards to reducing bill expenses are tremendous Green technology purifies the air, hence it will slow down the effects of global warming due to a reduction in CO_2 emissions. Generating electricity from solar and wind energy means less consumption of fossil fuels and a reduction in pollution and greenhouse gas emissions. Reducing and recycling of plastic waste is advantageous for the environment. Hence, trendy reusable water bottles that can be refilled are health-promoting, eco-friendly, and green.

vi. **Creates new avenues and Employment**. Green technologies have the potential to give birth to sectors that were previously not thought of. In this 21st century, we need to create paths that can improve the economy and the environment at the same time. Green Technology allows us to combine the two. We can see a very relevant example of that of waste disposal. Earlier waste management was only limited to waste dumping; today waste management. Is a $25 billion industry in South Asia alone? The green energy sector is responsible for a host of job opportunities on the market today. As a result, many employment options come up for people, and some of them include environmental managers, solar energy experts, and efficient lighting experts as well.

vii. **Recycling**. Green technology helps manage and recycle waste material. It allows it to be used for beneficial purposes. This technology is used for waste management, waste incineration, and more. A lot of recyclable material has allowed individuals to create plant fertilizer, sculptures, fuel, and even furniture.

viii. **Purifying of Water**. Green technology purifies water. The scarcity of pure drinking water is a major concern. Through the use of various technologies, a lot of campaigns have been successful in providing people with clean drinking water.

ix. **Green Farming**: Green approaches to farming have been proven to be not only healthier for humans but also productive for the soil. It leads to higher productivity over sustained periods contrary to the inorganic farming practices which lead to a decrease in yield after a certain period. Inorganic farming methods have had a very bad environmental impact and have resulted in degradation of the aquatic life of surface water bodies; it has also stripped the earth of the various insects and worms which helped in crop production. Fortunately, the effects of inorganic farming are yet to be seen in a big way but they are inevitable. Places, where organic (Green) farming is practiced, are already showing that it is a better approach in the long run.

x. **Green Buildings**: Green construction technologies are also coming up and are being encouraged in a big way. They have high initial investment but have a minimal environmental impact and are energy efficient. Because office and residential buildings consume a large share of the energy pie of any country they are certain to have significant

benefits in the future over conventional buildings. Since they reduce energy consumption and wastage; these buildings can recover their cost over an acceptable time frame. Such constructions prove economical and eco-friendly in the long run and thus are beneficial to the individual and the society as a whole

xi. **A benefit to the urban areas:** Taking into account the current chaotic situation of the cities of the world one can easily argue that they need to take urgent environment improvement measures. Cities that actively pursued their environmental concerns in the last ten years are showing a marked improvement in their environment quality parameters. For example, Delhi launched CNG fuelled public transport in a phased manner and in December 2002, the last diesel bus was flagged off. This was done as a measure to improve the air quality of Delhi where the toxic gas levels were off the charts, sometimes exceeding 5-12 times the normal values. Since then Delhi has shown steady improvement in the air quality. The annual average level of restorable suspended particulate matter (RSPM or PM10) in residential areas was 143 micrograms per cubic meter. It dropped to 115 micrograms per cubic meter by 2005.

xii. **Benefits to the Rural Areas: G**reen technologies involve humans in a much bigger way than conventional technologies and thereby empower them by giving them responsibilities and avenues to gain, learn and progress. Green technologies have had a great impact on communities of the areas where they have been implemented. The provision of biogas plants to rural households has empowered communities and has increased their productivity. The same has been the case with the distribution of solar lanterns through certain programs. Similarly, Green technology requires more involvement. Hence it empowers people. It (green technologies) helps people as it can be diffused much more easily in remote areas due to its discretized nature.

WEAKNESSES / DISADVANTAGES OF GREEN TECHNOLOGY

Green technologies have been applied in many sectors however they have not been put into complete use e.g. in the energy sector they are still “alternate sources of energy”. Since we have not seen them in usage full time we cannot observe what are their disadvantages. However certain concerns have been already raised, these are related to the reliability of these technologies, convenience of their usage, the investment required, their hypocritical use, ethical and social issues, etc. We will discuss these issues one by one.

i. **The initial investment or implementation cost is very high.** For instance; India's upper limit of the installation cost of a 1 MW photovoltaic solar power plant is roughly Rs 300-350 million (discounting the government grants). On the other hand, the installation cost of a subcritical coal power plant (Ultra mega power plant 4000MW) is roughly Rs 184,736 million. i.e. Rs 46.184 million per MW. The difference is extremely large and quite unaffordable for many developing countries; large investments in green technologies by the government would effectively slow down the cash flow in other important sectors e.g. health care and infrastructure.

ii. **Social downside**: Socially green technologies are still to become popular. This is more connected to the technical assistance and incentives of usage. Social downsides are extremely area specific. For instance in rural areas; people lack expertise, they also lack the proximity of servicing as well as sensitization and awareness creation opportunities. For urban areas, sometimes; the solar water heaters fail to work due to week insolation like in winter seasons hence people opt for electrical heaters which are always active as long as there is power supply constantly.

iii. **The technology is still evolving and many of the products are at the R&D stage**. Hence people are unaware of performance results. For instance; today solar panels are being manufactured at a prolific rate, however, it is not certain what will happen to the panels once they finish their life cycle. The disposal of solar panels is still in the R&D stage. Many solar panels contain toxic matter which can be dangerous; this problem will surface after two decades if we are not prepared by then.

iv. **Technical Downside:** To be popular any technology has to be reliable. Some green technologies have severe drawbacks in this area. For instance; solar power plants work fine on a sunny day but their performance becomes miserable the moment solar radiation on the surface drops, this cannot be afforded in today's competitive world. Similarly, solar air conditioning fails in the rainy season when there is a lot of humidity and low solar radiation. These drawbacks push a person to install conventional technologies for his/her usage. We can hope that these flaws will be removed once these technologies mature with time. Power savers meant to improve the efficiencies of tube lights etc. themselves fail very early giving the use of little incentive to use them. This causes significant operational discomfort to the user.

v. **Ethical Downside:** Today the world is in a very complicated state where ethics have started clashing, many times environmental ethics clash with general human ethics. A very relevant example is when USA started to use corn to make ethanol for bio-diesel. How ethical is it to use food to create fuel for cars when millions are hungry no matter how environment friendly the technology is? This argument effectively stalled this project. Green buildings, solar power plants all are currently very expensive; how justified is their implementation when it is more necessary to feed a hungry population? Can a poor person be forced to purchase new fuel efficient kitchen equipment when he barely makes his ends meet? There are umpteen ethical issues that spring up and are mostly related to cost versus survival questions. Prices of green technologies will go down in the future, but the basic problem is that their prices will go down once we make the initial investment but doing the initial investment is not easy. This is a vicious circle and is perhaps the biggest difficulty in the implementation of green technologies.

vi. **Land use:** A lot of green energy sources use more land than other non-green sources, which is a big negative if land use, space available, and arable land are important (depending on the location). For instance, A 12 KW home generator is only like 28 inches x 26 inches…a

similar sized solar panel would only generate like 100 watts (0.1 KW) (Alice, What is your view on the conversion of land from food production into energy crop production?

vii. **Green energy is still focused on fixed base consumption**. By this I mean its common applications are residential and commercial power. Green technology in the transportation sector is less mature. Although this is changing rapidly, battery technology still has a ways to go down the technology evolution path. Fuel cells are a potential solution, but the energy tradeoffs to generate the hydrogen for the cells still need some efficiency improvements. If it comes in greater availability it will allow greater ranges for electric vehicles.

Along with any energy solution(fossil or green) that we use in the future, we will require a 'smart grid' or electrical grid that allows the more efficient transmission of power over longer distances, as well as, storage of excess power during periods of low demand. With a smart grid, any limitations of generation points vs. consumption points disappear. This is still lacking hence hindering green technology's applicability.

We might end up seeing many effects that we cannot think as of now because we still have not experienced green technologies in a big way. We cannot say for certain that green technologies will be good for us in all ways in the distant future, but one thing is for certain that we will be risking the fate of mankind if we do not adopt them in the present. Today green technologies are the need of the hour without any doubt as they are most appropriate to our current world needs.

GREENING IN AGRICULTURE AND ENVIRONMENT

Greening it activities such as protecting permanent grassland designated as environmentally sensitive grassland, growing a minimum number of crops. Farming five per cent of your arable area in a manner that promotes biodiversity, i.e. Ecological Focus Area.

The aims of greening

The farmers act as managers of the countryside, they shape landscapes and through their work, farmers provide public goods beneficial to all. The "green direct payment" (or "greening") supports farmers who adopt or maintain farming practices that help meet environmental and climate goals.

Green economy

In a green economy, growth in employment and income are driven by public and private investment into such economic activities, infrastructure and assets that allow reduced carbon emissions and pollution, enhanced energy and resource efficiency, and prevention of the loss of biodiversity and ecosystem services. The benefits of green economy in businesses are; enhanced brand image and increased competitive advantage, increased productivity and reduced costs, better financial and investment opportunities, increased preparedness for future legislation and costs, improved recruitment and retention of quality employees, and a healthier work environment for employees.

RENEWABLE ENERGY

Renewable energy comes from sources or processes that are constantly replenished. These sources of energy include solar energy, wind energy, geothermal energy, and hydroelectric power.

Renewable sources are often associated with green energy and clean energy, but there are some subtle differences between these three energy types. Where renewable sources are those that are recyclable, clean energy are those that do not release pollutants like carbon dioxide, and green energy is that which comes from natural sources. While there is often cross-over between these energy types, not all types of renewable energy are actually fully clean or green. For example, some hydroelectric sources can actually damage natural habitats and cause deforestation.

TYPES OF RENEWABLE ENERGY

There are a range of renewable energy sources that have been developed, with each offering their own advantages and challenges depending on factors such as geographical location, requirements for use and even the time of year.

1. Solar Power

The potential for the sun to supply our power needs is huge, considering the fact that enough energy to meet the planet's power needs for an entire year reaches the earth from the sun in just one hour. However, the challenge has always remained in how to harness and use this vast potential.

We currently use solar energy to heat buildings, warm water and power our devices. The power is collected using solar, or photovoltaic (PV), cells made from silicon or other materials. These cells transform sunlight into electricity and can power anything from the smallest garden light to entire neighborhoods'. Rooftop panels can provide power to a home, while community projects and solar farms that use mirrors to concentrate the sunlight can create much larger supplies. Solar farms can also be created in bodies of water, called 'floatovoltaics' these provide another option for locating solar panels. As well as being renewable, solar powered energy systems are also clean energy sources, since they don't produce air pollutants or greenhouse gases. If the panels are responsibly sited and manufactured they can also count as green energy as they don't have an adverse environmental impact.

2. Wind Power

Wind energy works much like old-fashioned windmills did, by using the power of the wind to turn a blade. Where the motion of these blades would once cause millstones to grind together to make flour, today's turbines power a generator, which produces electricity.

When wind turbines are sited on land they need to be placed in areas with high winds, such as hilltops or open fields and plains. Offshore wind power has been developing for decades with wind farms providing a good solution for energy generation while avoiding many of the complaints around them being unsightly or noisy on land. Of course, offshore use has its own drawbacks due to the aggressive environments the turbines need to operate in.

3. Hydroelectric Power

Hydroelectric power works in a similar manner to wind power in that it is used to spin a generator's turbine blades to create electricity. Hydro power uses fast moving water in rivers or from waterfalls to spin the turbine blades and is widely used in some countries. It is currently the largest renewable energy source in the United States, although wind energy is fast closing the gap.

Hydroelectric dams are a renewable energy source, but these are not necessarily green energy sources. Many of the larger 'mega-dams' divert natural water sources, which creates a negative impact for animal and human populations due to restricted access to the water source. However, if carefully managed, smaller hydroelectric power plants (under 40 megawatts) do not have such a catastrophic effects on the local environment as the divert just a fraction of the water flow.

4. Biomass Energy

Biomass energy uses organic material from plants and animals, including crops, trees, and waste wood. This biomass is burned to create heat which powers a steam turbine and generates electricity. While biomass can be renewable if it is sustainably sourced, there are many instances where this is neither green nor clean energy.

Studies have shown that biomass from forests can produce higher carbon emissions than fossil fuels, while also have an adverse impact on biodiversity. Despite this, some forms of biomass do offer a low-carbon option given the correct circumstances. Sawdust and wood chippings from sawmills, for example, can be used for biomass energy where it would normally decompose and release higher levels of carbon into the atmosphere.

6. Geothermal

Geothermal energy uses the heat trapped in the Earth's core which is created by the slow decay of radioactive particles in rocks at the centre of the planet. By drilling wells, we are able to bring highly heated water to the surface which can be used as a hydrothermal resource to turn turbines and create electricity. This renewable resource can be made greener by pumping the steam and hot water back into the earth, thereby lowering emissions.

The availability of geothermal energy is closely tied to geographical location, with places such as Iceland having an easily reached, ready supply of geothermal resources.

7. Tidal Power

Tidal power offers a renewable power supply option, since the tide is ruled by the constant gravitational pull of the moon. The power that can be generated by the tide may not be constant, but it is reliable, making this relatively new resource an attractive option for many.

However, care needs to be taken with regard to the environmental impact of tidal power, as tidal barrages and other dam-like structures can harm wildlife.

Advantages of renewable energy

Renewable energy offers a range of benefits including offering a freely available source of energy generation. As the sector grows there has also been a surge in job creation to develop and install the renewable energy solutions of tomorrow. Renewable sources also offer greater energy access in developing nations and can reduce energy bills too.

Of course, one of the largest benefits of renewable energy is that much of it also counts as green and clean energy. This has created a growth in renewable energy, with wind and solar being particularly prevalent.

However, these green benefits are not the sole preserve of renewable energy sources. Nuclear power is also a zero-carbon energy source, since it generates or emits very low levels of CO_2. Some favour nuclear energy over resources such as solar and wind, since nuclear power is a stable

source that is not reliant on weather conditions. Which brings us onto some of the disadvantages of renewable energy.

Disadvantages of renewable energy

As mentioned above, many renewable energy sources cannot be relied upon all the time. When the sun goes down or hides behind a cloud, we cannot generate solar power and when the wind doesn't blow, we cannot create enough wind energy. For this reason, fossil fuels are still in use to top-up renewable sources in many countries.

This variable production capacity means that large energy storage solutions are required to ensure there is enough power when renewable energy generation dips. An alternative solution is to deploy several renewable technologies, creating a more flexible system of supply that can counteract dips in production for a given source.

Some renewable resources, such as hydropower and biomass, do not suffer with these problems of supply, but these both have their own challenges related to environmental impact, as noted above. In addition to this, some renewable energy sources, such as solar and wind farms, create complaints from local people who do not want to live near them.

However, this is not always the case, as shown by the example of Ardossan Wind Farm in Scotland, where most local residents believe the farm enhanced the area. Furthermore, a study by the UK Government found that, "projects are generally more likely to succeed if they have broad public support and the consent of local communities. This means giving communities both a say and a stake." This theory has been proven in Germany and Denmark, where community-owned renewable projects have proven popular.

BUILT ENVIRONMENT

The term built environment refers to the human-made surroundings that provide the setting for human activity, ranging in scale from buildings and parks or green space to neighborhoods and cities that can often include their supporting infrastructure, such as water supply or energy networks.

The built environment is a material, spatial, and cultural product of human labor that combines physical elements and energy in forms for living, working, and playing. It has been defined as "the human-made space in which people live, work, and recreate on a day-to-day basis"

A built environment is developed in order to satisfy residents' requirements. Human needs can be physiological or social and are related to security, respect, and self-expression. People want their built environment to be aesthetically attractive and to be in an accessible place with a well-developed infrastructure, convenient communication access, and good roads, and the dwelling should also be comparatively cheap, comfortable, with low maintenance costs, and have sound and thermal insulation of walls.

People are also interested in ecologically clean and almost noiseless environments, with sufficient options for relaxation, shopping, fast access to work or other destinations, and good relationships with neighbors.

The study of the built environment is interdisciplinary in nature and can include such disciplines as:

- visual arts,
- architecture,
- engineering,
- urban planning,
- history,
- interior design,
- industrial design,
- geography,
- environmental studies,
- anthropology / sociology

It must be admitted that the most serious problems of built environments (eg, unemployment, vandalism, lack of education, robberies) are not always related to the direct physical structure of housing. Increasing investment into the development of social and recreational centers, such as athletic clubs, physical fitness centers, and family entertainment centers, the infrastructure, a good neighborhood and better education of young people, can solve such problems. Investment, purchase and sale of a property, and its registration have related legal issues. The legal system of a country aims to reflect its existing social, economic, political, and technical state and the requirements of the market economy. As illustrated, the life cycle of the built environment can be assessed taking into account many quantitative and qualitative criteria. Life cycle of the built environment quantitative and qualitative analyses aspects are presented in Fig. 1.

Fig. 1. Life cycle of the built environment quantitative and qualitative analyses aspects.

Each one of these Level 1 aspects subsystems (see Fig. 1) based on the principle of a tree diagram can be discussed in much greater detail. To illustrate, the life cycle of the built environment described according to an example of the ith level aspects subsystem could be energy. In view of the global practice, any analysis of various aspects characteristic to the life cycle of the built environment focuses on the analysis of energy. Life cycle of the energy-efficient built environment application areas are presented in Fig. 2.

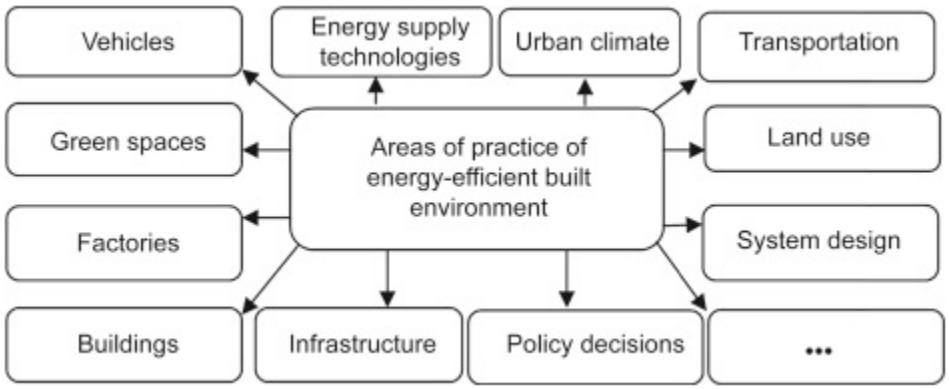

Fig. 2. Areas of practice of energy-efficient built environment.

Here is an overview of the applications used in the built environment:

- ***Domotic and home automation:*** Energy and water use (energy and water supply consumption monitoring to obtain advice on how to save cost and resources), remote control appliances (switching on and off appliances remotely to avoid accidents and save energy), intrusion detection systems (detection of window and door openings and violations to prevent intruders), art and goods preservation (monitoring of conditions inside museums and art warehouses).
- ***Smart cities:*** Smart parking (monitoring of parking spaces availability in the city), structural health (monitoring of vibrations and material conditions in buildings, bridges, and historical monuments), noise urban maps (sound monitoring in bar areas and centric zones in real-time), electromagnetic field levels (measurement of the energy radiated by cell stations and WiFi routers), traffic congestion (monitoring of vehicles and pedestrian levels to optimize driving and walking routes), smart lighting (intelligent and weather-adaptive lighting in street lights), waste management (detection of rubbish levels in containers to optimize the trash collection routes), smart roads (intelligent highways with warning messages and diversions according to climate conditions and unexpected events like accidents or traffic jams).
- ***Smart environment:*** Forest fire detection (monitoring of combustion gases and preemptive fire conditions to define alert zones), air pollution (control of CO_2 emissions of factories,

pollution emitted by cars), snow level monitoring (snow level measurement to know in real time the quality of ski tracks and allow security corps avalanche prevention), landslide and avalanche prevention (monitoring of soil moisture, vibrations, and earth density to detect dangerous patterns in land conditions), earthquake early detection (distributed control in specific places of tremors).

- ***Smart water:*** Potable water monitoring (monitor the quality of tap water in cities),
  chemical leakage detection in rivers (detect leakages and wastes of factories in rivers), swimming pool remote measurement (control remotely the swimming pool conditions), pollution levels in the sea (control real-time leakages and wastes in the sea), water leakages (detection of liquid presence outside tanks and pressure variations along pipes), river floods (monitoring of water level variations in rivers, dams, and reservoirs).
- ***Smart metering:*** Smart grid (energy consumption monitoring and management), tank level (monitoring of water, oil, and gas levels in storage tanks and
  cisterns), photovoltaic installations (monitoring and optimization of performance in solar energy plants), water flow (measurement of water pressure in water transportation systems), silos stock calculation (measurement of emptiness level and weight of the goods).
- ***Security and emergencies:*** Perimeter access control (access control to restricted areas and detection of people in non-authorized areas), liquid presence (liquid detection in data centers, warehouses, and sensitive building grounds to prevent breakdowns and corrosion),
  radiation levels (distributed measurement of radiation levels in nuclear power stations surroundings to generate leakage alerts), explosive and hazardous gases (detection of gas levels and leakages in industrial environments, surroundings of chemical factories, and inside mines).
- ***Retail:*** Supply-chain control (monitoring of storage conditions along the supply chain and product tracking for traceability purposes), NFC payment (payment processing based on
  location or activity duration for public transport, gyms, theme parks, etc.), intelligent shopping applications (getting advice in the point of sale according to customer habits, preferences, presence of allergic components for them, or expiring dates), smart product management (control of rotation of products in shelves and warehouses to automate restocking processes).